

Interesting Reference Designs Of GaN ADAPTORS/CHARGERS

Title of Reference Design	GaN-Based High Power Density Adaptor	GaN-Based QR Adaptor
Image		
Key Attractions	The reference design provides power efficiency, a small layout, and modular flexibility. It supports wide input voltage, reduces energy loss, improves cooling, and simplifies testing and upgrades with interchangeable modules.	The power adaptor reference design achieves over 93.6% efficiency, meets EMI and efficiency standards, supports USB-C, and maintains stable thermal performance under full load for reliable operation.
Highlights	The NCP13992UHD300WGEVB from Onsemi is a GaN-based adaptor reference design that helps engineers build small, efficient power solutions. It includes a power factor correction (PFC) stage and an LLC power stage, both using GaN transistors for high efficiency. The PFC stage, controlled by the NCP1616, improves power quality, while the LLC stage, managed by the NCP13992, runs at 500kHz for better performance. GaN Systems' transistors are used on the primary side, and the secondary side has an NCP4306-controlled synchronous rectifier with two MOSFETs per branch. The design achieves a power density of 32W/in ³ with a simple two-layer PCB, delivering up to 300W at 19V output. It supports a wide input range and meets CoC5 Tier 2 standards. The modular design, with a main board and plug-in modules, makes testing and upgrades easier while saving PCB space. The two-layer PCBs with thick copper improve cooling and reduce energy loss, especially in high-current areas.	The GS-EVB-ADP-65WQR-GS1 by GaN Systems uses a 650V, 150mΩ GaN transistor for power conversion, achieving over 93.6% peak efficiency and a power density of 18.5W/in ³ in a cased design. It meets EMI EN55032 Class B standards for conducted and radiated emissions and exceeds CoC Ver5 Tier 2 requirements for efficiency and standby power consumption. The circuit architecture includes the NCP1342 quasi-resonant (QR) controller on the primary side and a 100V Si MOSFET with the NCP4306 synchronous rectification (SR) controller on the secondary side, along with a power delivery (PD) protocol controller for USB-C compatibility. The design maintains EMI margins of 10dB for conducted and 6dB for radiated emissions at 230V and 110V AC inputs. After 30 minutes of full-load operation at 25°C ambient temperature, the PCB temperature remains below 100°C, and the cased adaptor's surface does not exceed 66°C.
Applications	The reference design is suitable for laptop and notebook adaptors.	The reference design suits USB-C adaptors for fast chargers, industrial controllers, and IoT power supplies.
OEM Brand	Onsemi	GaN Systems
URL	https://www.electronicsforu.com/news/gan-based-high-power-density-adaptor-reference-design	https://www.electronicsforu.com/electronics-projects/gan-based-qr-adaptor-reference-design